CNSCC.405, Lab 5

AWGN 和无线衰落信道数字调制方案的 BER 评估（Rayleigh & Rician）

Evaluation of BER of Digital Modulation Schemes for AWGN and Wireless fading Channels (Rayleigh & Rician)

随附的研究文章用于执行此实验室：

The research article is attached to perform this Lab:

Tasks: 1.

Simulate the BER of Digital Modulation Schemes for AWGN and Wireless Fading Channels (Rayleigh & Rician) with the guidelines in the attached article. (Make any realistic assumptions if required). You are required to simulate at least any two Digital Schemes (not even present in this article), but you can simulate more if you want because our more focus is on Fading channels rather digital modulation.

Compare the simulated results with the theoretical results (use “erfc” function for theoretical BER).

Tasks: 2.

Write a summary of what you have learned, observed, things went wrong (if in your experiment, you have not had the predicted result), and how you can overcome the similar problem in future.

Task 1:
根据所附文章中的指南，模拟 AWGN 和无线衰落通道（瑞利和莱斯）的数字调制方案的 BER。（如果需要，做出任何现实的假设）。您需要模拟至少任何两个数字方案（本文中甚至没有介绍），但如果您愿意，您可以模拟更多，因为我们更关注的是衰落通道而不是数字调制。

将仿真结果与理论结果进行比较（使用“erfc”函数计算理论 BER）

Task 2:
写下你所学到的、观察到的、出现问题的总结（如果在你的实验中，你没有得到预测的结果），以及你将来如何克服类似的问题。

BER是“比特误码率”（Bit Error Rate）的缩写，是通信领域用来衡量数字信号传输质量的一个指标。它表示在一定时间内，传输的数字信号中错误的比特数占总传输比特数的比例。

每比特信噪比
(SNR指标)

E_b/N_0 是一个非常重要的参数，用来衡量信号质量
[每比特能量与单位带宽噪声功率的比值]

E_b 表示 “每比特的能量”（bit energy），指传输一个比特数据所消耗的能量。这个能量越大，信号就越强，通常接收的正确性也越高。

N_0 表示“噪声功率谱密度”（noise power spectral density），即每单位带宽上的噪声功率，描述了噪声的强度。

Evaluation of BER of Digital Modulation Schemes for AWGN And Wireless Fading Channels

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Abstract— With the growing development of digital communication systems, it has became necessary to ensure the quality of service for real-time transmission of video applications and give users more powerful and efficient services by using better techniques. Digital modulation techniques have various performance characteristics power efficiency, Bandwidth efficiency, Error rate, cost etc. The M-ary Phase Shift Keying and M-ary Quadrature Amplitude Modulation are used in wireless communication to check test the performance of fading channels.
In this paper, evaluation of BER is performed under AWGN, Rayleigh and Rician fading channels. Among these channels, Rician has showing better performances a compared to AWGN and Rayleigh channels.

Keywords—Fading; AWGN; Rayleigh; Rician; MPSK; MQAM

I. INTRODUCTION

During the last few years we have witnessed a massive growth and commercial success of mobile communication systems and wireless network. The next generation of wireless communication systems will face the demand for increased data rates, higher mobility, heigher carrier frequency, and link reliability [1]. The performance of digital communication systems are related to the characteristics of the wireless propagation channel. It is therefore necessary to have an understanding of the behavior of wireless channels to determine their performance limits.

A thorough analysis of the propagation channel is necessary for strategic planning and system design by making comparative study of different modulation techniques via different multipath communication channels [2]. this paper considers the analysis of the performance of digital communications systems with different modulation schemes. In wireless communication, the performance of data transmission is captured by the checking their bit error rate (BER), which is a function of signal to noise ratio (SNR) at the receiver. Several models have been studied to calculate SNR [2][3]. These models use a path loss exponent, and a channel gain. Several distributed probability functions are available to model the wireless propagation channel. We describe the three major frequently used distributions. Such those named Additive White Gaussian noise (AWGN), Rayleigh and Rice models, … [2-5].

This paper has been organized as follows: In section II, different types of fading channel have been described. MARY modulations techniques are presented in section III. The mathematical analysis is presented in section IV. Simulation results are described in section V. Finally, some conclusions have been drawn in section VI.

II. FADING

In wireless communications, fading is considered as the deflection or attenuation induced by a propagation media. Fading may vary with time, geographical position and radio frequencies. Fading may either be due to multipath propagation, referred to as multipath induced fading, or due to shadowing from obstacles affecting the wave propagation, sometimes referred to as shadow fading. In this paper, we are interested in two types of fading due to multipath:

- Rayleigh Fading,

- Rician Fading.

A. Rayleigh Fading

In urban crossing buildings , vehicles and other objects, the transmitted, Tx, signals arrive at the receiver, Rx, on different paths as presented in Fig.1. A Rayleigh channel is considered when there are different signal paths between the transmitter and receiver, none of which dominates. This means that all the paths can vary and affect the overall signal at the receiver.

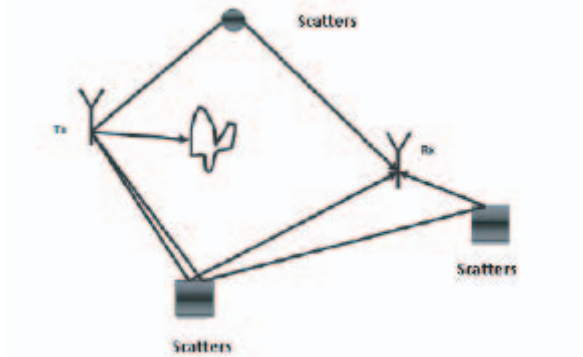

Figure.1 Rayleigh Channel Model

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The probability density function of a random variable is defined by [3-6]:

$$P_R(r) = \frac{r}{\sigma^2}\exp(-\frac{r^2}{2\sigma^2}) \text{ pour } r \geq 0 \quad (1)$$

Where r is the envelope amplitude of Rx Signal, σ^2 is the variance of the random variable.

Figure 2 shows the probability density for a fading Rayleigh channel for two values of variance σ^2.

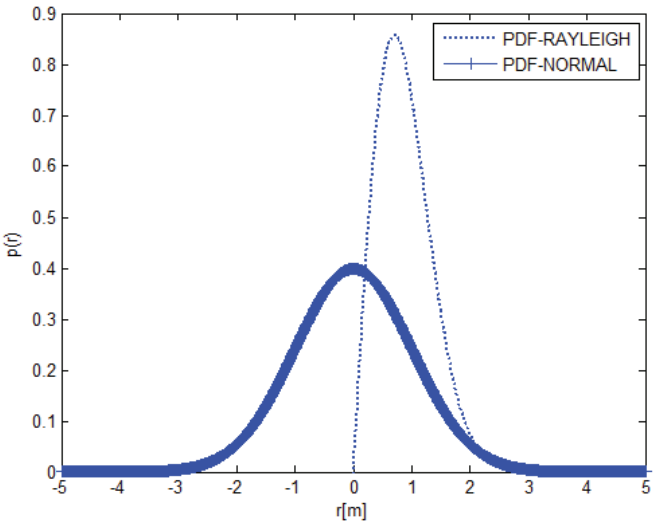

Figure.2 Probability density function of a fading channel Rayleigh

The graph (Figure 2) shows that the Rayleigh distribution does not present a symmetry as the normal distribution.

B. Rician Fading

A fading Rician channel occurs when the received signal is a combination of a line of direct view and multipath fading between a transmitter and a receiver as shown in Fig.3. The line of sight path is the strongest signal path that travels directly from the transmitter to receiver. Due to direct path, the effect of Rician fading on the transmitted signal will be less than in the case of Rayleigh fading [6].

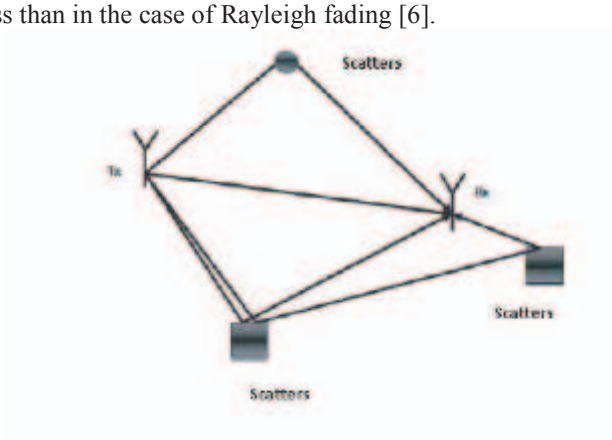

Figure.3 Rician Channel Model

The probability density function of a random variable is [7][8]:

$$P_R(r) = \frac{r}{\sigma^2} e^{-(\frac{(r^2+s^2)}{2\sigma^2})} I_0(\frac{rs}{\sigma^2}) \text{ for } r \geq 0, s \geq 0 \quad (2)$$

Where σ^2 is the average power in the NLOS (Non Line Of Sight) multipath components and s^2 is the power in the LOS (Line Of Sight) component [9]. The function I_0 is the zeroth-order modified Bessel function of the first kind it is defined by the eq. 3 as shown in [6][7][8].

$$I_0(x) \triangleq \frac{1}{\pi}\int_0^{\pi} \exp(x\cos\theta)d\theta \quad (3)$$

Expression (2) can be written as follows:

$$P_R(r) = \frac{r}{\sigma^2} e^{-K} e^{-\frac{r^2}{2\sigma^2}} I_0(\frac{r\sqrt{2K}}{\sigma}) \quad (4)$$

Where the factor K is one of the inputs that define the ratio between the power of the LOS component and multipath components obtained as [1]:

$$K = \frac{s^2}{2\sigma^2} \quad (5)$$

Figure 4 shows the probability density of the Rician fading channel for different values of the direct component, s , and for $\sigma^2 = 0.5$.

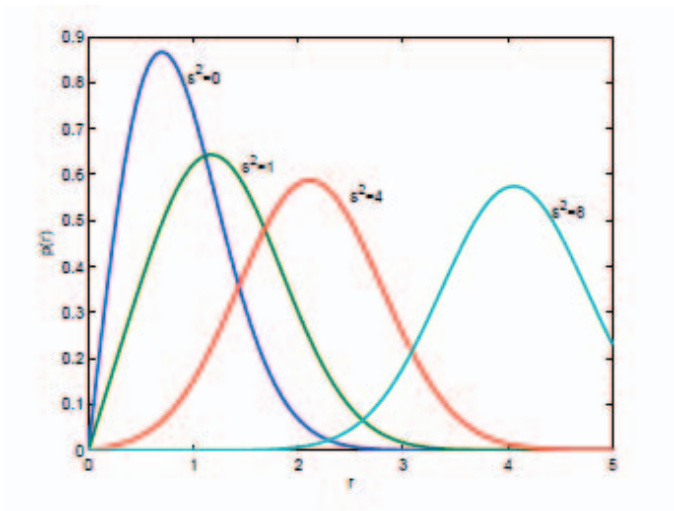

Figure.4 Probability density function of a fading channel Rician

When the direct component is zero, we have Rayleigh Fading, while when it increases; we have no fading, i.e. a channel with no multipath and only a LOS component.

III. M-ARY MODULATION TECHNIQUE

A. M-QAM modulation scheme

In digital communication, data are first transformed to binary form, i.e. the baseband data amplitude can only take 0 and 1 values. The baseband data is first applied to a serial to parallel convertor to generate two sets data with half data rate each.

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These two sets of data will modulate two orthogonal signals with the same carrier frequency to generate In-phase (I) and Quadrature (Q) components. I- and Qsignals are then added to form the final modulated signal. This is called quadrature amplitude modulation (QAM). MQAM, Mary Quadrature Amplitude Modulation is widely used in modulation technique with respect to many features like: high transmission rates, high spectral efficiency, correct configuration and high energy efficiency as shown in [10-11]. The most common forms are 16- QAM, 64-QAM and 256-QAM , which transmit 4, 6, and 8 bits per symbol respectively *(fig 5)*.

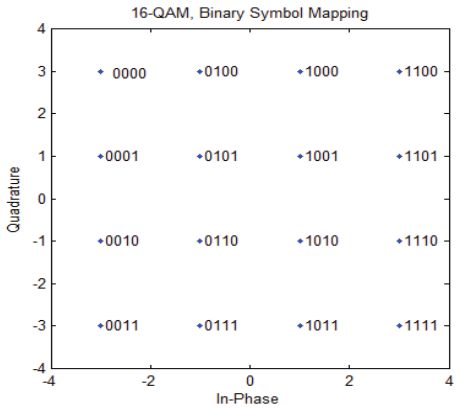

Figure.5 Constellation 16QAM

B. M-PSK modulation scheme

M-ary Phase Shift Keying (M-PSK) is a modulation scheme that conveys data by changing, or modulating, the phase of a reference signal. A finite number of phases are used to represent digital data. PSK is widely used in the communication industry [7]. A variety of forms of PSK are available and some forms are used in this paper binary PSK (Binary phaseshift keying, BPSK), quadrature PSK (Quadrature phase-shift keying, QPSK) and 8PSK.

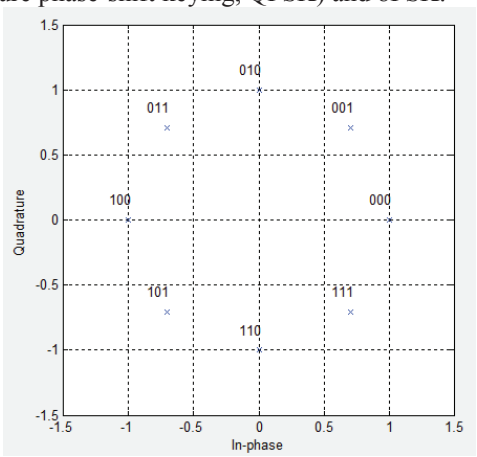

Figure.6 Constellation 8PSK

IV. BER ANALYSIS

In digital modulation techniques, due to noise, interference, and distortion the received bits are altered .So bit error rate is defined as the number of error bits over total number of transmitted [6]. The bit error probability is the mean value of the BER as shown in [12].

$$P_b = \frac{Number \text{ of bits in error}}{Total\ Number \text{ of transfered bits}} \tag{6}$$

The bit error rate of coherent digital modulation is generally given as [13]:

$$P_b(E \mid \gamma) = aQ(\sqrt{b\gamma}) \tag{7}$$

Where γ is the SNR per bit is the ratio between the energy per information bit Eb and the variance of the two dimensional noise N0 $\gamma = E_b / N_0$, 'a' and 'b' are certain values depending on the modulation schemes. Q is the function to calculate the area under the tail of the Gaussian; denoted by Q(x) [13].

$$Q(x) = \frac{1}{\pi}\int_0^{\frac{\pi}{2}} \exp\left(-\frac{x^2}{2\sin^2\theta}\right) d\theta \tag{8}$$

We are interested in average probability of error performance over fading channel which can be obtained as [11-13]:

$$P_b(E) = \int_0^{\infty} aQ(\sqrt{b\gamma})p_\gamma(\gamma)d\gamma \tag{9}$$

Where $p_\gamma(\gamma)$ is the probability density function of SNR (γ).

The parameters 'a' and 'b' used are presented in Table 1 as indicated in [13].

Table 1 Parameters 'a' and 'b'

Modulation	a	b
MPSK	$\frac{1}{\log_2(M)}$	$2\log_2(M)\sin\frac{\pi}{M}$
MQAM	$\frac{4}{\log_2(M)}(1-\frac{1}{\sqrt{M}})$	$\frac{3}{M-1}\log_2(M)$

V. SIMULATION RESULT

In this section, the study of four digital modulation schemes i.e. 2-PSK, 8-PSK, 16-QAM and 64-QAM is performed on Additive White Gaussian Noise (AWGN), Line of Sight (Rician) and Non Line of Sight (Rayleigh) Fading Channels. The results are plotted using MATLAB in terms of bit error rate.

Fig. 7 presents a comparative study of AWGN channel, Rayleigh and Rician fading channels MPSK modulation (M = 2 and 8).

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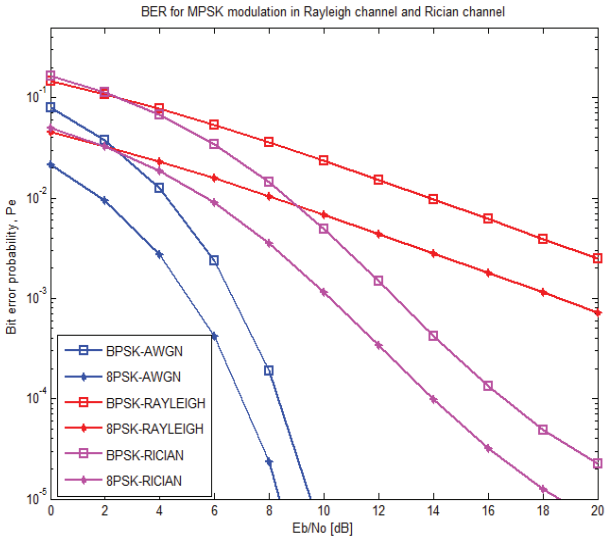

Figure.7 Bit error rate (BER) performance of AWGN, Raleigh and Rician channels for MPSK modulation technique

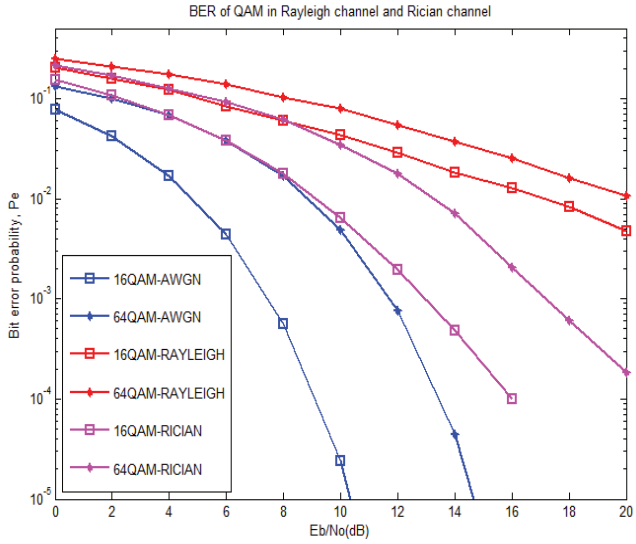

Figure.8 Bit error rate (BER) performance of AWGN, Raleigh and Rician channels for MQAM modulation technique

Table 2 BER in AWGN channel, Rayleigh and Rician fading channels in MPSK modulation scheme

Type Channel		MPSK	
		2	4
AWGN	E_b / N_0	BER	
	2	0.03750	0.00941
	10	0.00000	0.00040
	14	0.00000	0.00000
	18	0.00000	0.00000
Rayleigh	E_b / N_0		
	2	0.10848	0.03277
	10	0.02326	0.00681
	14	0.00966	0.00281
	18	0.00391	0.00113
Rician	E_b / N_0		
	2	0.11279	0.03275
	10	0.00491	0.00115
	14	0.00042	0.00009
	18	0.00004	0.00001

Table 3 BER in AWGN channel, Rayleigh and Rician fading channels in MQAM modulation scheme

Type Channel		MQAM	
		16	64
AWGN	E_b / N_0	BER	
	2	0.0417	0.0995
	10	0.0000	0.0049
	14	0.0000	0.0000
	18	0.0000	0.0000
Rayleigh	E_b / N_0	BER	
	2	0.1576	0.2088
	10	0.0424	0.0802
	14	0.0190	0.0385
	18	0.0076	0.0153
Rician	E_b / N_0	BER	
	2	0.1096	0.1661
	10	0.0060	0.0360
	14	0.0003	0.0066
	18	0.0000	0.0004

Fig. 8 presents a comparative study of AWGN channel, Rayleigh and Rician fading channels using MQAM modulation (M = 16 and 64).

From Fig. 7, 8 and tables 2 and 3, for large values of E_b/N_0, the probability of error decreases exponentially compared with E_b/N_0 in AWGN channel. While in a fading channel (Rayleigh, Rician), the probability of error decreases linearly. We also note that as the value of M, i.e., the number of bits in a symbol increases, the error rate also increases. So we deduce that the BER is minimum for AWGN and is maximum for Rayleigh and Rice. AWGN channel is not a fading channel, so BER vs E_b/N_0 will be compared with Rayleigh channel and Rice. Since the fading has a strong rice dominant component (LOS) is present, so that the BER value for the rice will certainly lower than the Rayleigh channel for the direct transmission path.

Taking the MQAM modulation, different E_b/N_0 values with a constant $BER = 10^{-2}$ are obtained by using different values of M as shown in Table 5. Under 16QAM modulation

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scheme, E_b/N_0 for AWGN channel remains 5 dB, where E_b/N_0 for Rayleigh and Rician channel remains 17 dB and 9 dB respectively. Using the 64QAM modulation scheme, for $BER = 10^{-2}$, E_b/N_0 for AWGN channel equals 8.8 dB, where E_b/N_0 for Rayleigh and Rician channel remains 20 dB and 13 dB respectively. Due to the absence of the direct line LOS from the transmitter to the receiver, Rayleigh channel has low performance compared to AWGNT and Rice channel.

Table 4 BER VS. E_b/N_0 COMPARISON FOR AWGN, RAYLEIGH AND RICIAN CHANNEL USING MPSK MODULATION.

Values of M	E_b/N_0		
	AWGN	**Rayleigh**	**Rician**
2	4.3	14	8.5
8	1.8	8.3	5.9

Table 5 BER VS. E_b/N_0 COMPARISON FOR AWGN, RAYLEIGH AND RICIAN CHANNEL USING MPSK MODULATION.

Values of M	E_b/N_0		
	AWGN	**Rayleigh**	**Rician**
16	5	17	9
64	8.8	20	13

VI. CONCLUSION

In the wireless communication system, the data rate requirements are increasing day by day to ensure application service quality that allows the real-time transmission of video. Several modulation techniques are used and by which it is possible to transmit more than the bit rate in the channel. In this paper, the evaluation of digital modulations (MPSK, MQAM) is performed for wireless applications in the AWGN channel and fading channels (Rayleigh and Rice). From the simulation results, we see that for the same values of M, Rice presents better performance than AWGN channel and Rayleigh fading channel constant in terms of BER with E_b/N_0. Because Rician fading channel has higher BER than AWGN channel and lower than Rayleigh fading channel. BER of this channel has not been much affected by noise under 2PSK, 8PSK, 16QAM & 64 QAM modulation schemes.

在 E_b/N_0 的 BER 方面，Rice 表现出比 AWGN 信道和瑞利衰落信道常数更好的性能。因为莱斯衰落信道的BER比AWGN信道高，比瑞利衰落信道低。该通道的 BER 在 2PSK、8PSK、16QAM 和 64 QAM 调制方案下受噪声影响不大。

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